In this paper, we investigate structure of group of small order.

Let G be an any group, and denote $|G|$ as order of G.

There are many tools to investigate the finite groups

「Not necessary to Consider the order.
- Lagrange's Theorem.
- Normalizer, Centralizer.

「Necessary order Consideration
- Class equation
- Sylow Theorem.

$\ulcorner |G| = 1$. It is just trivial.

$\ulcorner |G| = 2$. Let $G = \{1, a\}$. If $a^2 \neq 1$, then $a^2 = a$, therefore $1 \cdot a = a$ and $a \cdot a = a$, it means a has no inverse element. So, $a^2 = 1$, it means $G \cong \mathbb{Z}_2$.

$\ulcorner |G| = 3$. Let $G = \{1, a, b\}$.